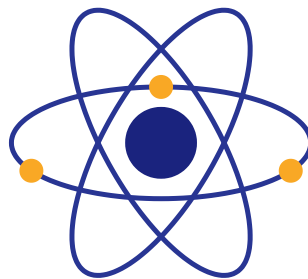


Basic Electrical & Electronics Engineering

BEEE | Regulation: R23 | Semester I

Prepared as per JNTUK R23 Syllabus | Exam-Oriented Study Material

UNIT – IV : Semiconductor Devices



Topics Covered in This Unit:

1. Introduction & Evolution of Electronics
2. Semiconductor Basics
3. PN Junction Diode – Formation & Characteristics
4. Forward Bias & Reverse Bias
5. VI Characteristics (Diode)
6. Zener Diode – Effect, Characteristics & Voltage Regulation
7. Bipolar Junction Transistor (BJT) – NPN & PNP
8. BJT Configurations – CB, CE, CC
9. Transistor Input & Output Characteristics
10. CE Amplifier – Working & Analysis
11. Comparison Tables
12. Questions & Answers (2M & 5M)

 **Exam Focus:** All major topics include neat labeled SVG diagrams, step-by-step explanations, key formulas, and structured answers suitable for scoring maximum marks.

Section 1: Introduction to Electronics

1.1 What is Electronics?

Electronics is the branch of engineering and science that deals with the flow and control of electrons in vacuum, gases, or semiconductor materials. It involves the design and application of circuits using components like diodes, transistors, and integrated circuits to process signals, control power, and perform logic operations.

Electronics forms the backbone of modern civilization — from communication systems and computers to medical instruments and consumer gadgets. Without electronics, technologies such as radio, television, mobile communication, the internet, and computers would not exist.

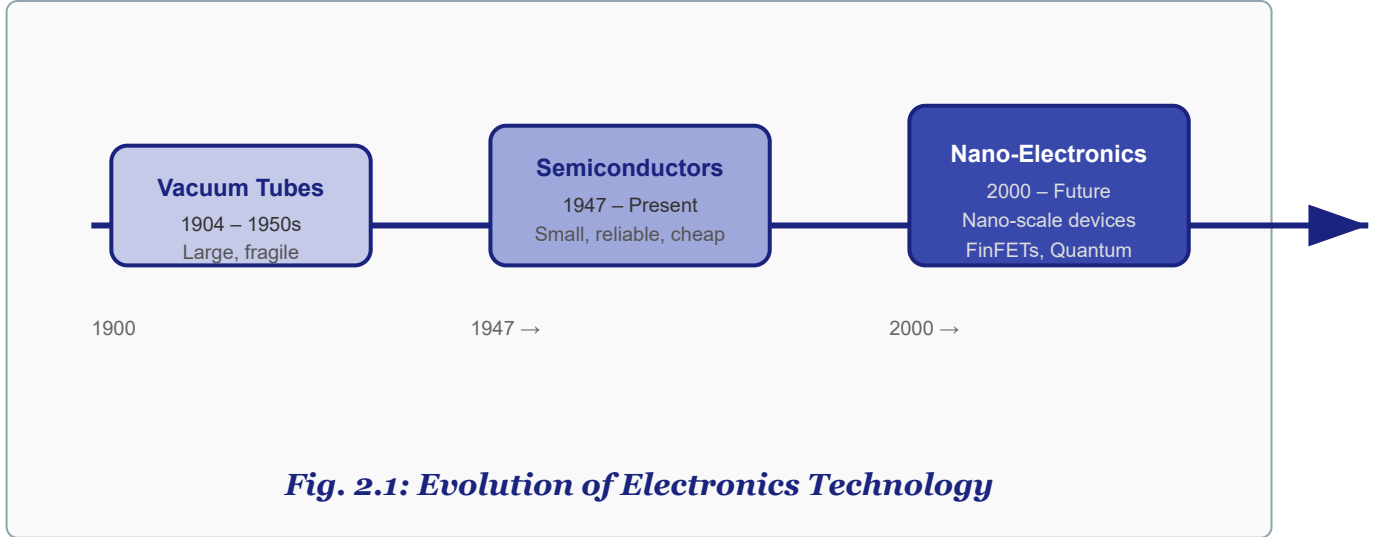
1.2 Branches of Electronics

- **Analog Electronics:** Deals with continuously variable signals (e.g., amplifiers, oscillators).
- **Digital Electronics:** Deals with discrete binary signals (e.g., logic gates, microprocessors).
- **Power Electronics:** Deals with high-power conversion and control (e.g., inverters, converters).
- **Microelectronics:** Deals with miniaturized electronic components and ICs.

Section 2: Evolution of Electronics

2.1 Historical Overview

The evolution of electronics spans over a century, progressing through three major technological eras: the Vacuum Tube Era, the Semiconductor Era, and the Nano-Electronics Era. Each era brought revolutionary improvements in size, speed, efficiency, and capability.



2.2 Vacuum Tube Era (1904 – 1950s)

The first electronic devices were vacuum tubes, invented by John Ambrose Fleming (1904). They were used for amplification and signal processing but were bulky, power-hungry, fragile, and generated enormous heat. Early computers like ENIAC (1945) used over 18,000 vacuum tubes and occupied entire rooms.

Key Devices: Diode tube, Triode, Tetrode, Pentode

Disadvantages: Large size, high power consumption, low reliability, slow warm-up time.

2.3 Semiconductor Era (1947 – Present)

The invention of the Bipolar Junction Transistor (BJT) by Shockley, Bardeen, and Brattain at Bell Labs in 1947 marked the dawn of the semiconductor era. Transistors replaced vacuum tubes, enabling miniaturization of circuits. The development of Integrated Circuits (ICs) by Jack Kilby (1958) and Robert Noyce (1959) revolutionized computing and communication.

Key Milestones: BJT (1947), IC (1958), MOSFET (1960), Microprocessor Intel 4004 (1971)

2.4 Nano-Electronics Era (2000 – Future)

Modern electronics operates at the nanometer scale. Transistors in contemporary processors (e.g., Apple M-series chips) have gate lengths below 3 nm. Research in quantum computing, carbon nanotube transistors, spintronics, and molecular electronics is paving the way for post-silicon computing.

Parameter	Vacuum Tube	Transistor	IC/Nano-Electronics
Size	Large (cm)	Small (mm)	Nano-scale (nm)
Power	Very High	Low	Ultra Low

Reliability	Poor	Good	Excellent
Cost	High	Low	Very Low per unit
Speed	Slow	Fast	Ultra Fast (GHz–THz)
Heat Generated	Enormous	Moderate	Minimal

Section 3: Semiconductor Basics

3.1 What is a Semiconductor?

A semiconductor is a material whose electrical conductivity lies between that of a conductor (like copper) and an insulator (like glass). At room temperature, semiconductors have a partially filled valence band and a narrow energy gap (~1 eV) between the valence band and conduction band. This allows limited but controllable electrical conduction.

Examples of Semiconductors: Silicon (Si), Germanium (Ge), Gallium Arsenide (GaAs), Indium Phosphide (InP)

3.2 Types of Semiconductors

A. Intrinsic Semiconductor

A pure semiconductor with no impurities is called an intrinsic semiconductor. At absolute zero (0 K), it behaves like an insulator. At room temperature, thermal energy breaks covalent bonds, generating electron-hole pairs. The number of free electrons (n) equals the number of holes (p).

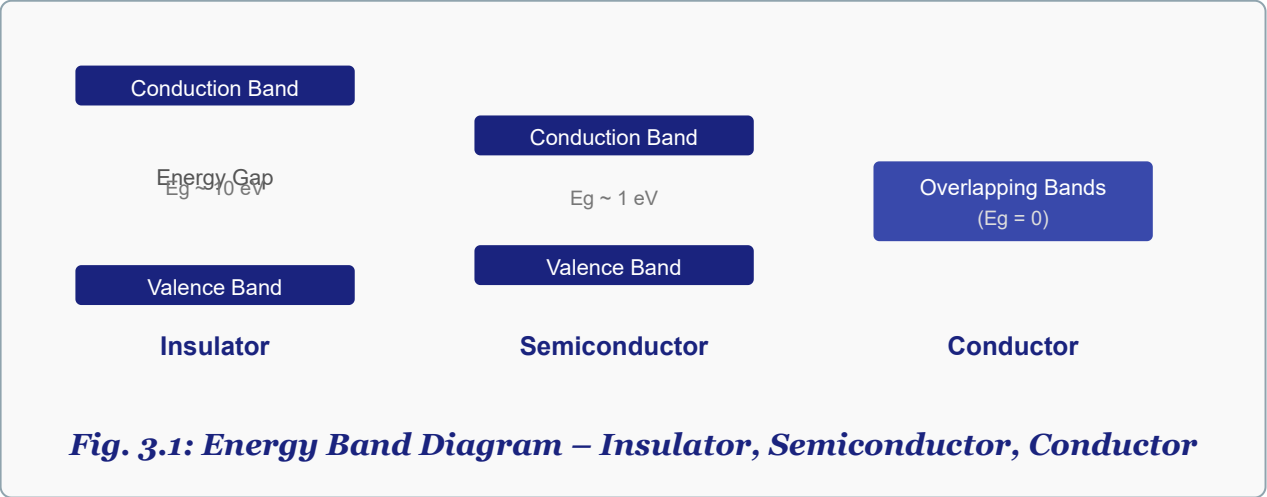
$$n_i = n = p \quad | \quad n_i = \sqrt{(N_c \cdot N_v) \cdot e^{-E_g/2kT}}$$

B. Extrinsic Semiconductor (Doped)

When specific impurities (dopants) are intentionally added to a pure semiconductor, it becomes an extrinsic semiconductor. Doping dramatically increases conductivity.

Type	Dopant	Majority Carrier	Minority Carrier	Example
N-type	Pentavalent (P, As, Sb)	Electrons	Holes	Si + Phosphorus
P-type	Trivalent (B, Al, Ga)	Holes	Electrons	Si + Boron

3.3 Energy Band Diagram



Section 4: PN Junction Diode

4.1 Definition

A PN junction diode is a two-terminal semiconductor device formed by joining a P-type semiconductor with an N-type semiconductor. It is the most fundamental semiconductor device and forms the basis of all advanced semiconductor devices like BJT, FET, thyristors, etc.

4.2 Formation of PN Junction

When a P-type and N-type semiconductor material are brought into contact, the following physical phenomena occur:

1. **Diffusion:** Free electrons from the N-region diffuse into the P-region due to the concentration gradient. Similarly, holes from the P-region diffuse into the N-region.
2. **Recombination:** Diffusing electrons recombine with holes near the junction, and diffusing holes recombine with electrons. This creates a region devoid of free charge carriers called the **Depletion Region**.
3. **Space Charge Formation:** Ionized donor atoms (positive) remain in the N-side and ionized acceptor atoms (negative) remain in the P-side, creating an electric field across the junction.
4. **Built-in Potential (V_{bi}):** The electric field creates a built-in potential barrier (~ 0.7 V for Si, ~ 0.3 V for Ge) that opposes further diffusion and establishes equilibrium.

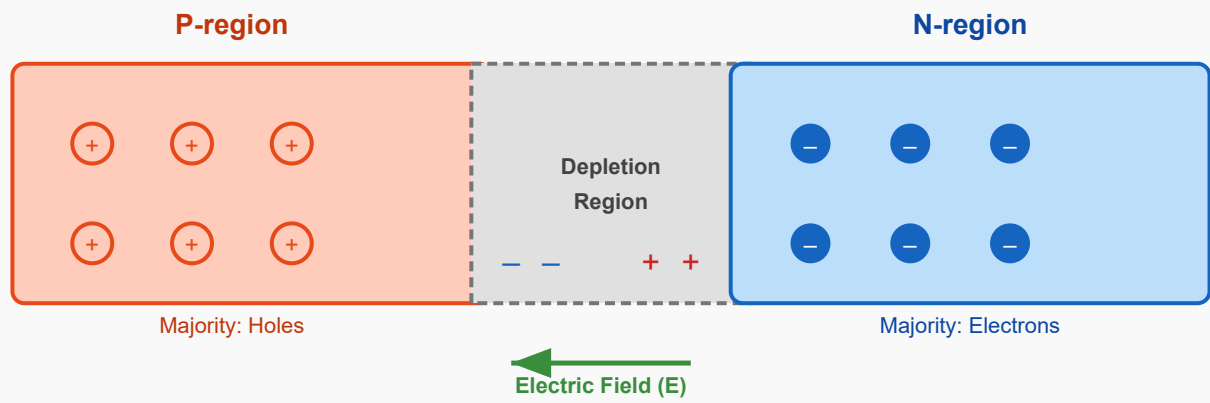


Fig. 4.1: PN Junction Diode – Structure showing P-region, N-region, and Depletion Region

4.3 Diode Symbol and Terminal Identification

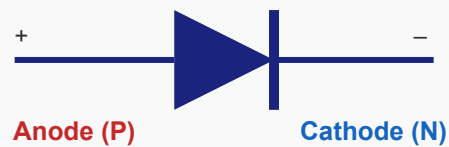


Fig. 4.2: PN Junction Diode Symbol

Diode Current Equation: $I = I_S (e^{V/\eta V_T} - 1)$

Where: I_S = Reverse Saturation Current | η = Ideality Factor (1 for Ge, 2 for Si) | V_T = Thermal Voltage = $kT/q \approx 26 \text{ mV}$ at 300K

Section 5: Diode Characteristics (VI Characteristics)

5.1 Forward Bias

When the positive terminal of a battery is connected to the P-type and the negative terminal to the N-type of a PN junction, the diode is said to be in forward bias.

- The external voltage opposes the built-in potential, reducing the depletion region width.
- When the forward voltage exceeds the cut-in voltage (0.7 V for Si, 0.3 V for Ge), the barrier is overcome and significant current flows.
- The forward resistance is very low (10–100 Ω).

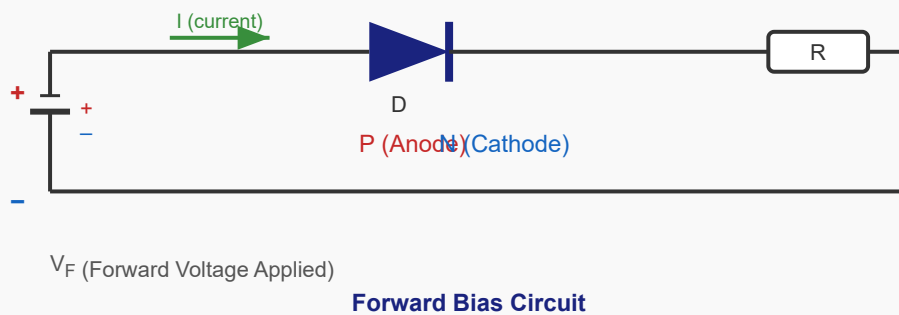
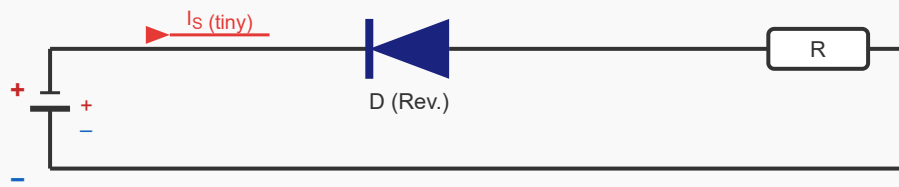


Fig. 5.1: PN Junction Diode – Forward Bias Circuit

5.2 Reverse Bias

When the positive terminal of a battery is connected to the N-type and the negative terminal to the P-type, the diode is in reverse bias.

- The external voltage adds to the built-in potential, widening the depletion region.
- Only a very small reverse saturation current (I_S) flows due to minority carriers (μA for Si).
- At a sufficiently high reverse voltage, breakdown occurs (Zener/Avalanche breakdown).



Reverse Bias Circuit

Fig. 5.2: PN Junction Diode – Reverse Bias Circuit

5.3 VI Characteristics (Voltage-Current Characteristics)

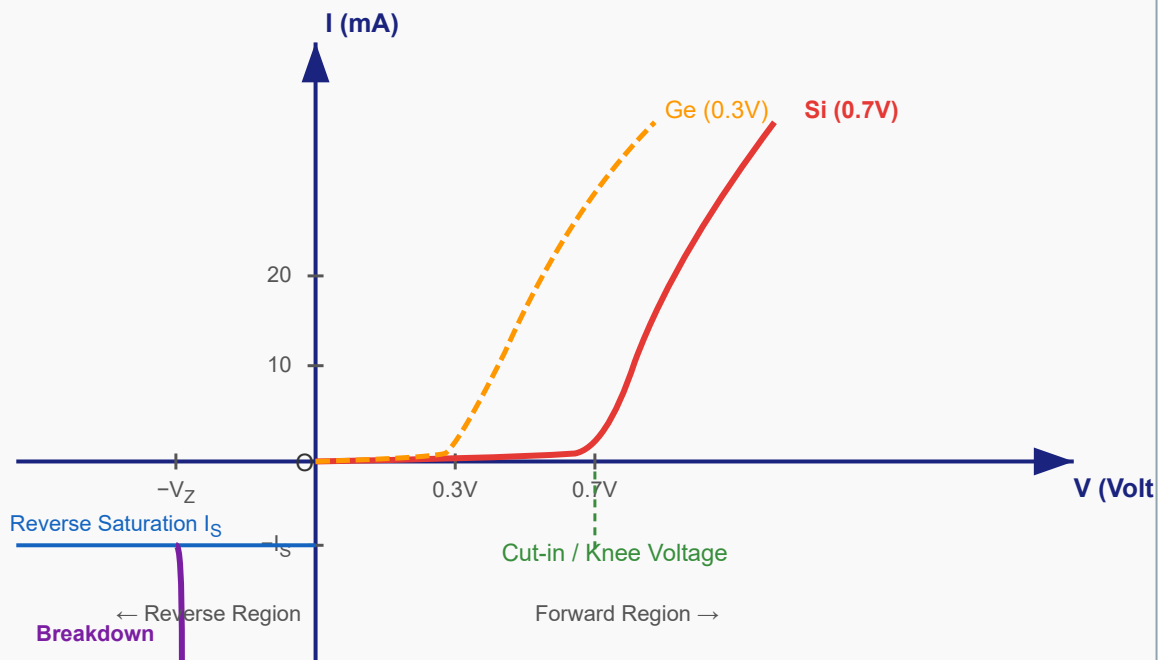


Fig. 5.3: VI Characteristics of PN Junction Diode (Si and Ge)

Key Observations from VI Characteristics:

1. In forward bias, current increases exponentially once $V > V_{\text{cut-in}}$.
2. In reverse bias, a small reverse saturation current flows (μA range).
3. At breakdown voltage (V_Z), reverse current increases sharply.

Section 6: Zener Diode

6.1 Definition

A Zener diode is a special type of PN junction diode that is designed to operate in the reverse breakdown region. Unlike ordinary diodes, a Zener diode maintains a nearly constant voltage across its terminals when reverse biased beyond its Zener voltage (V_Z), making it ideal for voltage regulation applications.

6.2 Zener Effect

The Zener effect (also called Quantum Mechanical Tunneling) occurs in heavily doped PN junctions with a thin depletion region. When a sufficiently large reverse voltage is applied, the electric field across the depletion region becomes intense enough to break covalent bonds and directly pull valence electrons across the junction without any collision, resulting in a sudden increase in reverse current at the Zener voltage (V_Z).

Two Types of Breakdown in Diodes:

- 1. Zener Breakdown:** Occurs in heavily doped diodes at $V_Z < 6V$. Due to strong electric field tunneling. Temperature coefficient is *negative*.
- 2. Avalanche Breakdown:** Occurs in lightly doped diodes at $V_Z > 6V$. Due to carrier multiplication by impact ionization. Temperature coefficient is *positive*.

6.3 Zener Diode Symbol and Construction

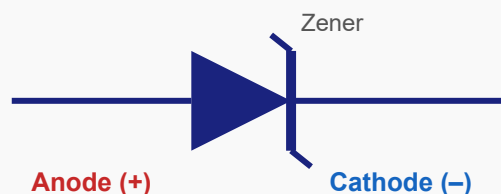


Fig. 6.1: Zener Diode Symbol (note bent cathode bar)

6.4 Zener Diode VI Characteristics

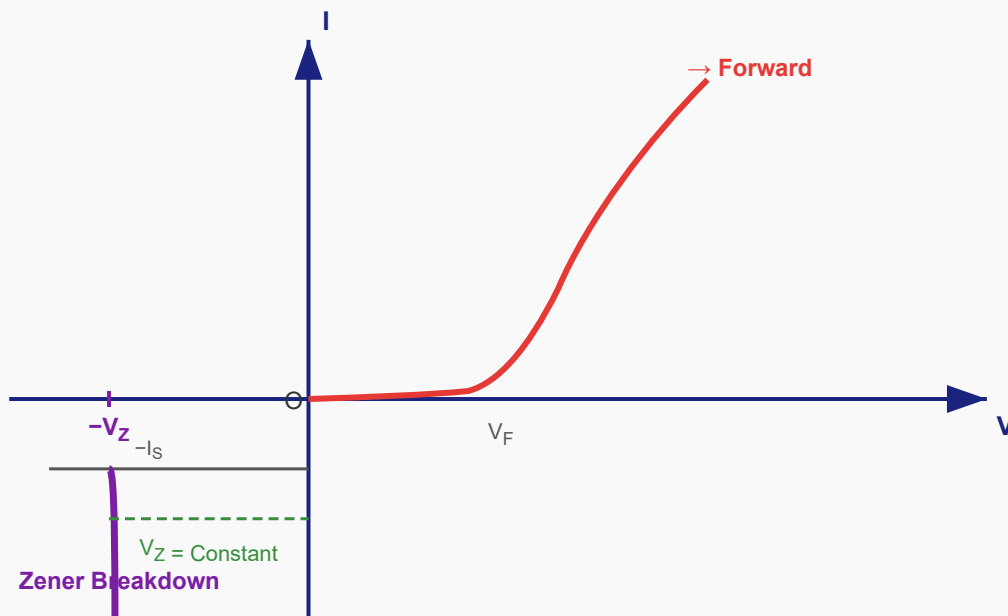


Fig. 6.2: Zener Diode VI Characteristics – showing Zener Breakdown Region

Zener Voltage: $V_Z = \text{Constant}$ (during breakdown) | Zener
Impedance: $r_Z = \Delta V_Z / \Delta I_Z$

Applications of Zener Diode:

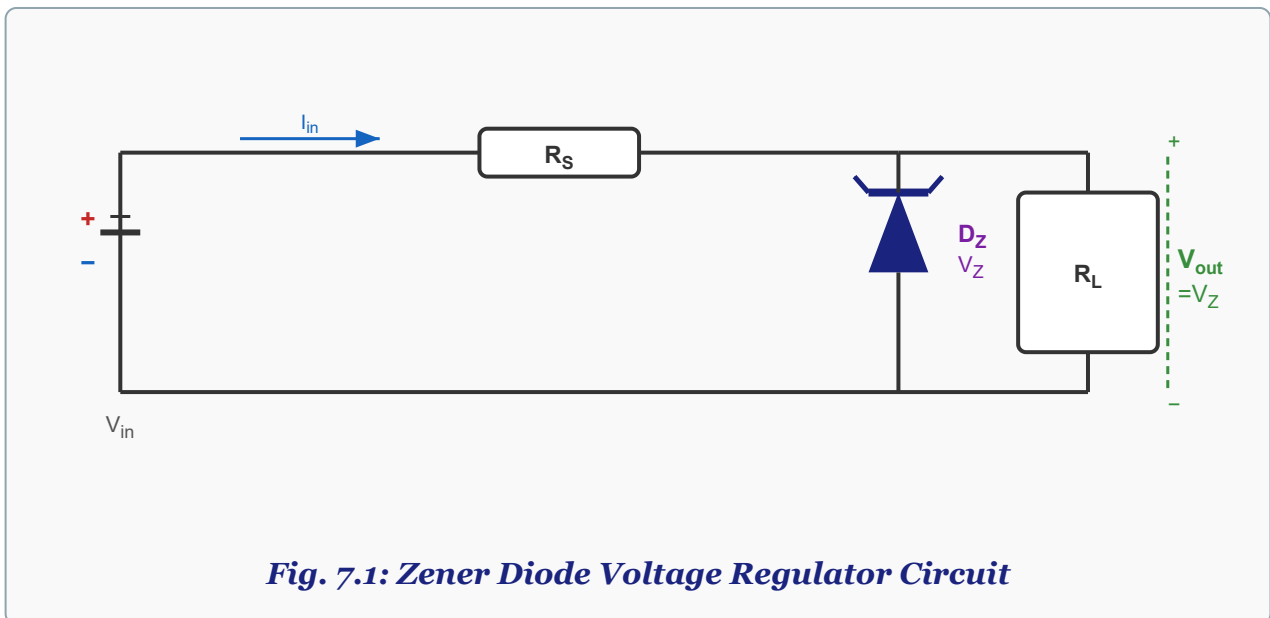
- Voltage regulation in DC power supplies
- Reference voltage in measurement circuits
- Over-voltage protection (surge protection)
- Waveform clipping/limiting circuits

Section 7: Zener Diode as a Voltage Regulator

7.1 Concept of Voltage Regulation

A voltage regulator is a circuit that maintains a constant output voltage despite variations in input supply voltage or load current. The Zener diode, when connected in reverse bias with a series resistor, acts as a shunt regulator — it maintains a constant voltage V_Z across the load even when the input voltage fluctuates.

7.2 Zener Voltage Regulator Circuit



7.3 Working Principle

1. The Zener diode is connected in reverse bias parallel to the load R_L .
2. The series resistor R_S limits the current through the circuit.
3. As long as the input voltage $V_{in} > V_Z$, the Zener diode conducts in breakdown and clamps the output to V_Z .
4. If V_{in} increases, the extra voltage drops across R_S , while V_{out} remains V_Z .
5. If the load current increases, Zener current decreases to compensate, maintaining $V_{out} = V_Z$.

$$\begin{aligned}V_{\text{out}} &= V_Z \quad | \quad I_S = (V_{\text{in}} - V_Z) / R_S \\I_S &= I_Z + I_L \quad | \quad I_L = V_Z / R_L \\R_{S(\text{min})} &= (V_{\text{in}(\text{max})} - V_Z) / (I_{Z(\text{max})} + I_{L(\text{min})})\end{aligned}$$

⚡ **Regulation Condition:** For the Zener diode to regulate effectively, I_Z must remain between $I_{Z(\text{min})}$ and $I_{Z(\text{max})}$ at all times.

Section 8: Bipolar Junction Transistor (BJT)

8.1 Introduction

A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device consisting of two back-to-back PN junctions. It is called "bipolar" because both electrons and holes participate in current conduction. The term "junction" refers to the two PN junctions, and "transistor" is derived from "Transfer Resistor" — a device that transfers current from a low resistance input to a high resistance output.

The BJT was invented by Shockley, Bardeen, and Brattain at Bell Laboratories in 1947. It is used for amplification, switching, and signal processing in electronic circuits.

8.2 Types of BJT

NPN Transistor

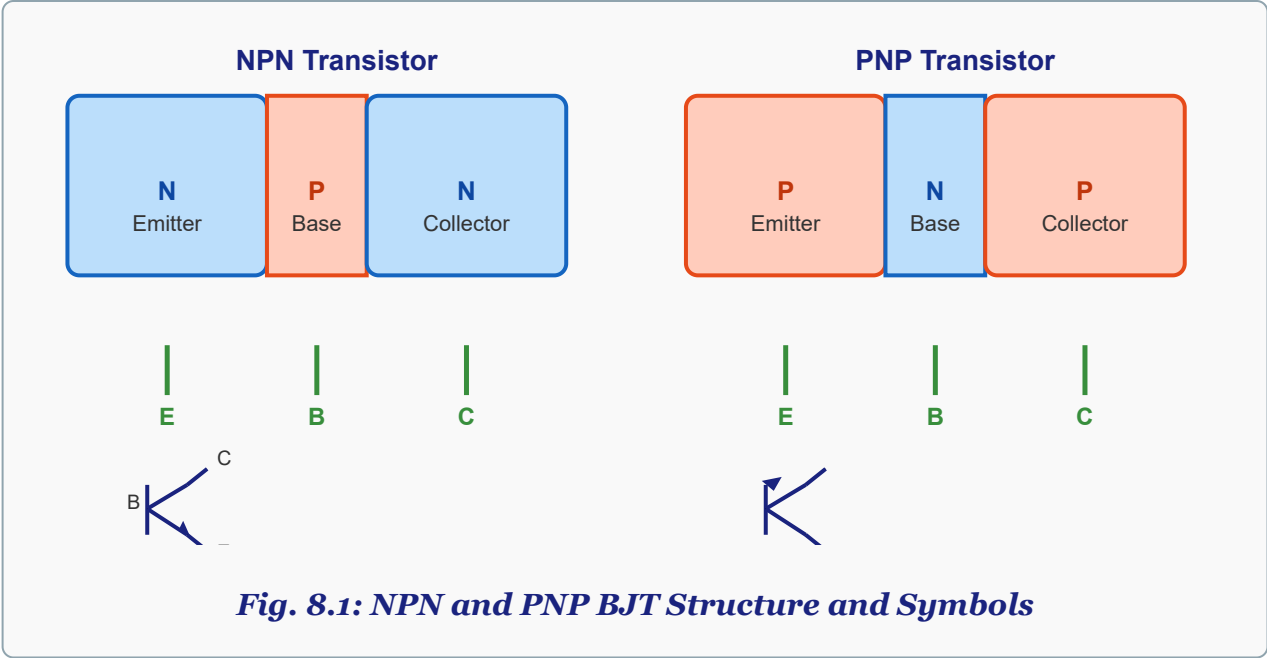
A thin P-type layer (base) is sandwiched between two N-type layers (emitter and collector).

Current Flow: Conventional current flows from Collector to Emitter (electrons from E to C).

PNP Transistor

A thin N-type layer (base) is sandwiched between two P-type layers (emitter and collector).

Current Flow: Conventional current flows from Emitter to Collector (holes from E to C).



8.3 BJT Terminals and Their Functions

Terminal	Function	Doping Level	Size
Emitter (E)	Emits/injects charge carriers into the base region	Heavily doped	Small
Base (B)	Controls current flow; acts as a gate	Very lightly doped	Very thin
Collector (C)	Collects charge carriers from the base region	Moderately doped	Largest

Fundamental BJT Relations:

$$I_E = I_B + I_C \quad | \quad \beta \text{ (} h_{FE} \text{)} = I_C / I_B \quad | \quad \alpha = I_C / I_E$$
$$\beta = \alpha / (1-\alpha) \quad | \quad \alpha = \beta / (\beta+1)$$

Typical values: $\beta = 20 \text{ to } 500 \quad | \quad \alpha \approx 0.95 \text{ to } 0.999$

Section 9: BJT Configurations

A transistor can be connected in three configurations depending on which terminal is common to both input and output ports. The three configurations are: **Common Base (CB)**, **Common Emitter (CE)**, and **Common Collector (CC)**.

9.1 Common Base (CB) Configuration

In this configuration, the Base terminal is common to both the input (Emitter-Base) and output (Collector-Base) circuits. The input signal is applied between Emitter and Base, and the output is taken from Collector and Base.

- **Current Gain (α):** Less than 1 (typically 0.95–0.999)
- **Voltage Gain:** High
- **Input Impedance:** Very Low (20–500 Ω)
- **Output Impedance:** Very High (M Ω)
- **Phase shift:** 0° (no inversion)
- **Application:** RF amplifiers, impedance matching

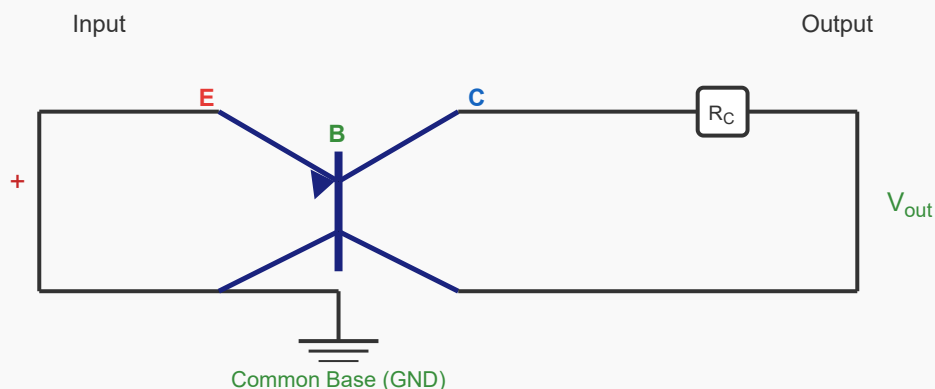


Fig. 9.1: Common Base (CB) Configuration of NPN Transistor

9.2 Common Emitter (CE) Configuration

In this configuration, the Emitter terminal is common to both input (Base-Emitter) and output (Collector-Emitter) circuits. This is the most widely used transistor configuration.

- **Current Gain (β):** 20 to 500 (highest among three)

- **Voltage Gain:** High
- **Power Gain:** Very High (highest)
- **Input Impedance:** Medium ($1\text{--}5\text{ k}\Omega$)
- **Output Impedance:** Medium ($20\text{--}50\text{ k}\Omega$)
- **Phase Shift:** 180° (output inverted)
- **Application:** Audio amplifiers, general purpose amplifiers

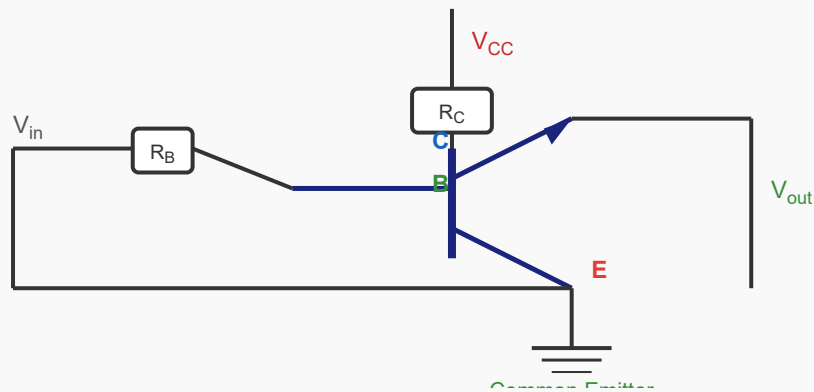


Fig. 9.2: Common Emitter (CE) Configuration of NPN Transistor

9.3 Common Collector (CC) Configuration (Emitter Follower)

In this configuration, the Collector terminal is common to both input and output. The output is taken from the Emitter terminal. It is also called an Emitter Follower because the output voltage follows the input voltage.

- **Current Gain:** $(\beta+1) \approx \beta$ (high)
- **Voltage Gain:** Less than 1 (≈ 1)
- **Input Impedance:** Very High ($100\text{ k}\Omega\text{--}1\text{ M}\Omega$)
- **Output Impedance:** Very Low ($25\text{--}100\text{ }\Omega$)
- **Phase Shift:** 0° (no inversion)
- **Application:** Buffer amplifier, impedance transformation, current amplification

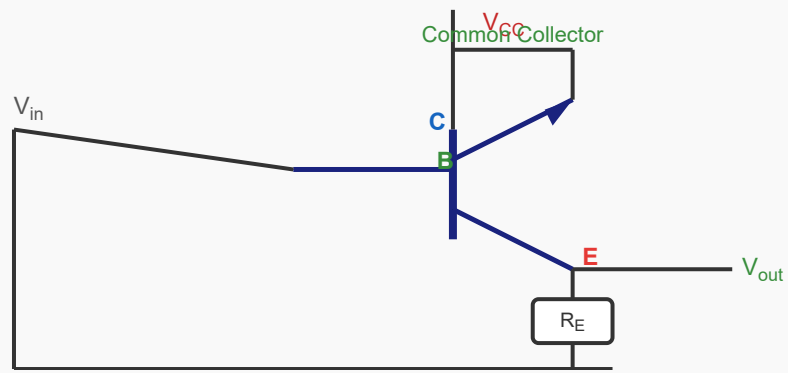


Fig. 9.3: Common Collector (CC) Configuration – Emitter Follower

Section 10: Transistor Characteristics (CE Configuration)

10.1 Input Characteristics (I_B vs V_{BE})

Input characteristics give the relationship between the base current (I_B) and the base-emitter voltage (V_{BE}) for a fixed collector-emitter voltage (V_{CE}). These characteristics resemble the forward VI characteristics of a PN junction diode.

Key Points:

- The curve is exponential, similar to a diode's forward characteristic.
- Cut-in voltage ≈ 0.7 V for Si.
- Input resistance: $h_{ie} = \Delta V_{BE} / \Delta I_B$
- Higher V_{CE} shifts the curve slightly to the right (Early Effect).

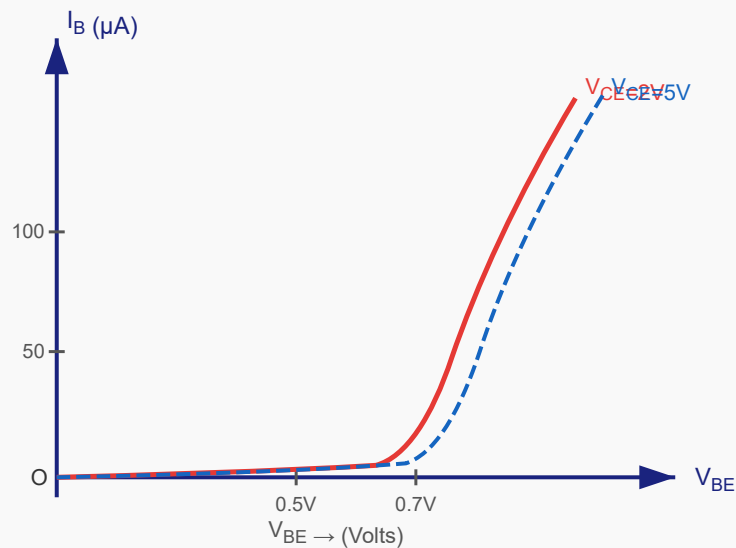


Fig. 10.1: CE Input Characteristics (I_B vs V_{BE})

10.2 Output Characteristics (I_C vs V_{CE})

Output characteristics show the relationship between Collector current (I_C) and Collector-Emitter voltage (V_{CE}) for different values of Base current (I_B).

Three distinct regions are identified in the output characteristics:

- **Active Region:** $I_C \approx \beta \cdot I_B$, used for linear amplification.

- **Saturation Region:** Both junctions forward biased, V_{CE} very small; transistor acts as a closed switch.
- **Cutoff Region:** Both junctions reverse biased, $I_C \approx 0$; transistor acts as an open switch.

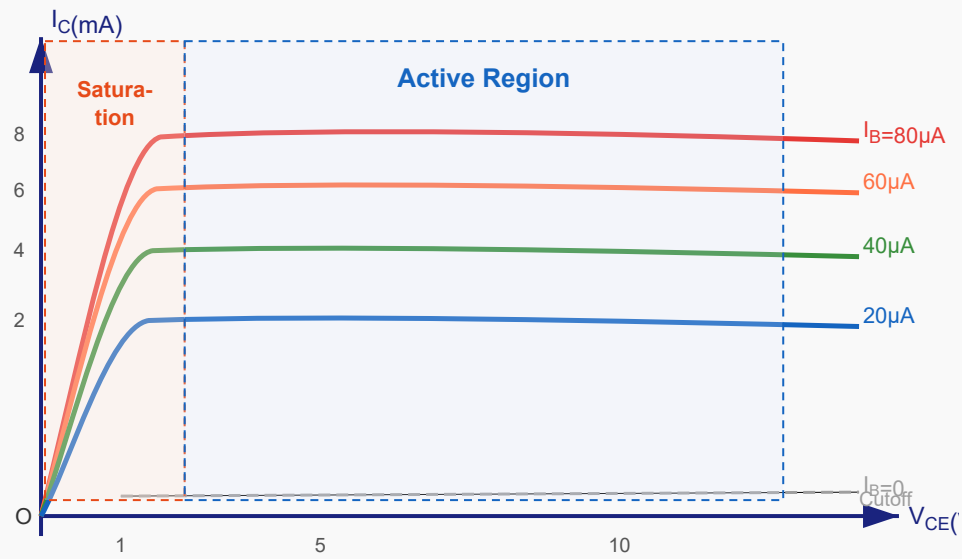


Fig. 10.2: CE Output Characteristics (I_C vs V_{CE}) showing Active, Saturation, and Cutoff regions

Section 11: Common Emitter (CE) Amplifier

11.1 Introduction

An amplifier is a circuit that increases the magnitude of an input signal. The Common Emitter amplifier is the most popular single-stage transistor amplifier because it provides both voltage gain and current gain, resulting in a high power gain. It is classified as a small-signal amplifier when the signal amplitude is small enough that the transistor operates in the linear region throughout the signal cycle.

11.2 CE Amplifier Circuit

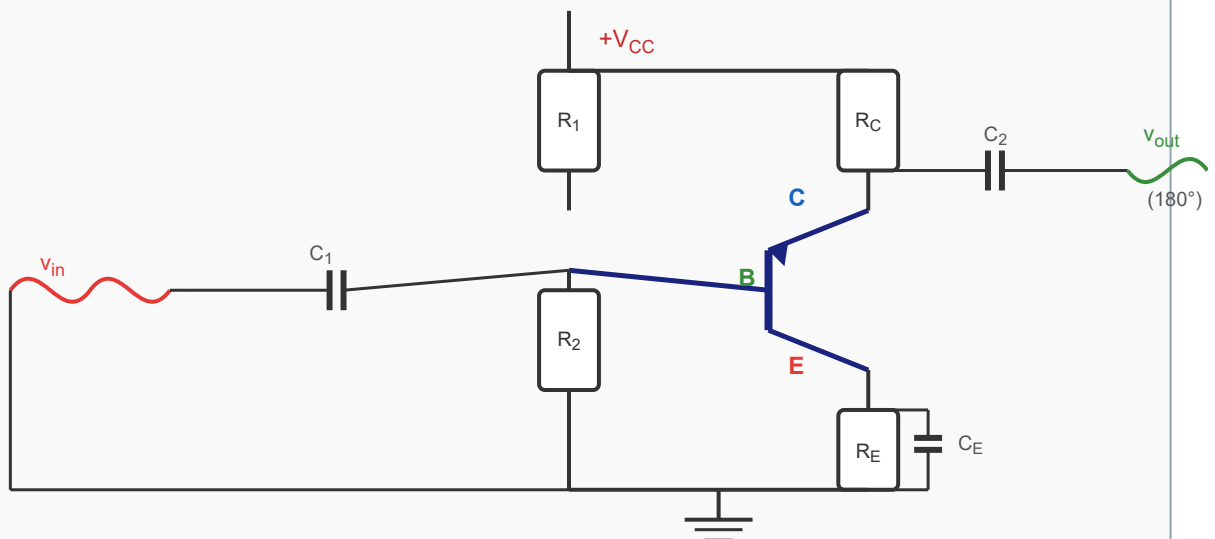


Fig. 11.1: Common Emitter Amplifier Circuit with Biasing Components

11.3 Working Principle (Step by Step)

- Biasing:** Resistors R_1 and R_2 form a voltage divider that sets the DC operating point (Q-point) in the active region of the transistor.
- Input Coupling:** C_1 is the input coupling capacitor that blocks DC and allows only AC signal to enter the base terminal.
- Signal Amplification:** The small AC signal at the base causes a proportionally large variation in collector current ($I_C = \beta \times I_B$), resulting in amplification.
- Output Voltage:** The amplified current through R_C produces a large voltage drop, and the output is taken across R_C (or the collector-emitter terminals).

5. **Phase Inversion:** When V_{in} increases (positive half cycle), I_B and I_C increase $\rightarrow$ more voltage drops across $R_C \rightarrow V_{CE}$ decreases $\rightarrow$ output goes negative. This causes 180° phase inversion.
6. **Bypass Capacitor C_E :** Bypasses the emitter resistor R_E for AC signals, increasing the AC voltage gain without disturbing the DC stability.
7. **Output Coupling:** C_2 blocks DC from the load and couples only the amplified AC signal to the output.

$$\text{AC Voltage Gain: } A_v = -\beta \cdot R_C / r_e \quad | \quad r_e = 26\text{mV} / I_C$$

$$\text{Power Gain: } A_p = A_v \times A_i \quad | \quad A_i = \beta$$

$$\text{Input Impedance: } Z_{in} = R_1 \parallel R_2 \parallel \beta \cdot r_e$$

Section 12: Comparison Tables

12.1 Comparison of BJT Configurations

Parameter	Common Base (CB)	Common Emitter (CE)	Common Collector (CC)
Input Terminal	Emitter	Base	Base
Output Terminal	Collector	Collector	Emitter
Common Terminal	Base	Emitter	Collector
Current Gain	$\alpha < 1$	$\beta = 20-500$	$\gamma = \beta + 1 \approx \beta$
Voltage Gain	High	High	$< 1 (\approx 1)$
Power Gain	Moderate	Highest	Moderate
Input Impedance	Very Low (20–500Ω)	Medium (1–5kΩ)	Very High (>100kΩ)
Output Impedance	Very High (MΩ)	Medium (20–50kΩ)	Very Low (25–100Ω)
Phase Shift	0° (no inversion)	180° (inverted)	0° (no inversion)
Applications	RF, high-freq amps	General amplifiers	Buffer/impedance matching

12.2 Comparison: PN Diode vs Zener Diode

Parameter	PN Junction Diode	Zener Diode
Normal Operation	Forward bias	Reverse bias (breakdown)
Symbol	Normal bar at cathode	Bent/Z-bar at cathode
Breakdown Voltage	High and not controlled	Precisely controlled (V_Z)

Doping	Light to moderate	Heavily doped
Main Application	Rectification	Voltage regulation
Operation in Breakdown	Destructive (damages diode)	Non-destructive (designed for it)
$V_{\text{cut-in}}$ (Si)	0.7 V	0.7 V (forward)

12.3 Comparison: NPN vs PNP Transistor

Parameter	NPN Transistor	PNP Transistor
Material Arrangement	N-P-N	P-N-P
Majority Carriers	Electrons	Holes
Current Direction (I_E)	Into Emitter ←	Out of Emitter →
V_{CE} Polarity	Positive	Negative
Speed	Faster (electrons more mobile)	Slower
Symbol Arrow (Emitter)	Pointing outward	Pointing inward
More Common?	Yes	Less common

12.4 Vacuum Tube vs Transistor vs IC

Feature	Vacuum Tube	Transistor	Integrated Circuit
Invented	1904	1947	1958
Size	Centimeter	Millimeter	Nanometer
Fragility	Very fragile (glass)	Rugged	Very rugged
Power	Watts–Kilowatts	Milliwatts–Watts	Microwatts–mW
Example	ENIAC computer	Early radios	Modern CPU/smartphone

Section 13: Questions & Answers

13.1 Two-Mark Questions

Q1. Define a PN junction diode. What is the depletion region?

2 Marks

PN Junction Diode: A PN junction diode is a two-terminal semiconductor device formed by joining a P-type material (rich in holes) and an N-type material (rich in free electrons). It allows current to flow primarily in one direction — from Anode (P) to Cathode (N) under forward bias — making it a fundamental rectifying device.

Depletion Region: When the P and N materials are joined, electrons from the N-side and holes from the P-side diffuse across the junction and recombine near the boundary. This creates a region depleted of free charge carriers (neither free electrons nor holes) known as the *depletion region* or *space charge region*. It acts as an insulating barrier and has an associated built-in electric field ($\sim 0.7\text{V}$ for Si) that opposes further diffusion.

Q2. What is the Zener effect? State any two applications of Zener diode.

2 Marks

Zener Effect: The Zener effect is a quantum mechanical tunneling phenomenon that occurs in heavily doped PN junction diodes when reverse biased beyond the Zener voltage (V_Z). Under the intense electric field across the thin depletion region, valence electrons gain enough energy to tunnel directly from the valence band to the conduction band without needing to cross the energy barrier conventionally. This results in a sharp increase in reverse current at V_Z while the voltage remains nearly constant.

Applications of Zener Diode:

1. *Voltage Regulation:* Maintains a constant output voltage in DC power supplies.
2. *Over-Voltage Protection:* Protects sensitive circuits from voltage spikes and surges.

Q3. Define α and β of a transistor. Write the relation between them.

2 Marks

α (Alpha) – Common Base Current Gain: α is defined as the ratio of Collector current to Emitter current in the Common Base configuration. It represents the fraction of emitter current that reaches the collector.

$$\alpha = I_C / I_E \text{ (always } < 1, \text{ typically } 0.95 - 0.999)$$

β (Beta) – Common Emitter Current Gain: β is defined as the ratio of Collector current to Base current in the CE configuration. It represents the amplification factor of the transistor.

$$\beta = I_C / I_B \text{ (ranges from 20 to 500)}$$

Relationship between α and β :

$$\beta = \alpha / (1 - \alpha) \text{ and } \alpha = \beta / (\beta + 1)$$

Since α is close to 1, β can be very large.

Q4. What is the difference between forward bias and reverse bias?

2 Marks

Forward Bias: When the positive terminal of the battery is connected to the P-type (Anode) and negative terminal to the N-type (Cathode) of a PN junction diode, it is said to be forward biased. In this condition, the depletion region narrows, the barrier potential reduces, and significant current flows when $V_F > V_{\text{cut-in}}$ (0.7V for Si). The diode resistance is very low in forward bias.

Reverse Bias: When the positive terminal is connected to the N-type and negative terminal to the P-type, the diode is reverse biased. The depletion region widens, the barrier increases, and only a negligible reverse saturation current (I_S , in μA) flows. The diode resistance is extremely high. At sufficiently high reverse voltage, breakdown occurs.

13.2 Five-Mark Questions

Q5. Explain the VI characteristics of a PN Junction diode with a neat diagram.

5 Marks

PN Junction Diode VI Characteristics:

The VI (Voltage-Current) characteristics of a PN junction diode describe how the current through the diode varies with the applied voltage. The diode is characterized by two distinct regions of operation: Forward Bias and Reverse Bias.

1. Forward Bias Characteristics:

When forward voltage is applied, initially (below cut-in voltage $\approx 0.7V$ for Si), the current remains very small because the applied voltage is insufficient to overcome the barrier potential. Beyond the cut-in (knee) voltage, the current increases exponentially with voltage, following the diode equation: $I = I_S(e^{V/\eta V_T} - 1)$. The dynamic forward resistance is very low.

2. Reverse Bias Characteristics:

Under reverse bias, only a very small reverse saturation current (I_S) flows, caused by minority carriers (typically in μA range for Si). This current is nearly constant and independent of the applied reverse voltage up to the breakdown voltage.

3. Breakdown Region:

At the Zener/avalanche breakdown voltage (V_Z), the reverse current increases sharply while the voltage across the diode remains approximately constant. This is a destructive breakdown for ordinary diodes but is the operating region for Zener diodes.

Diode Current Equation: $I = I_S(e^{V/\eta V_T} - 1)$

Where I_S = reverse saturation current, η = ideality factor, V_T = thermal voltage $\approx 26mV$ at $300K$.

[Refer to Fig. 5.3 in Section 5 for the VI characteristics diagram]

Q6. Describe the working of a Zener diode as a voltage regulator with a circuit diagram. 5 Marks

Zener Diode Voltage Regulator:

Circuit Description: A Zener diode voltage regulator consists of a series resistor R_S and a Zener diode D_Z (connected in reverse bias) in parallel with the load resistor R_L . The input voltage V_{in} must be greater than V_Z for proper regulation.

Working Principle:

When the input voltage $V_{in} > V_Z$, the Zener diode enters breakdown and clamps the output voltage at $V_{out} = V_Z$.

Case 1 – Input voltage increases: The increased voltage causes more current through R_S . The extra current flows through the Zener diode (I_Z increases), while the voltage across the Zener (V_Z) remains constant. The output voltage stays at V_Z .

Case 2 – Load current increases (R_L decreases): More current is demanded by the load. This is compensated by a decrease in Zener current (I_Z decreases), keeping the total current through R_S approximately constant and maintaining $V_{out} = V_Z$.

Key Formulas:

Total current: $I_S = I_Z + I_L$

Series resistor current: $I_S = (V_{in} - V_Z) / R_S$

Load current: $I_L = V_Z / R_L$

Output: $V_{out} = V_Z = \text{Constant}$

Condition for Regulation: $I_{Z(\min)} \leq I_Z \leq I_{Z(\max)}$ at all times.

[Refer to Fig. 7.1 in Section 7 for the circuit diagram]

Q7. Explain the three configurations of BJT with their characteristics and applications.

5 Marks

BJT Configurations:

A transistor has three configurations based on which terminal is common to input and output:

1. Common Base (CB): Input at Emitter, output at Collector, Base is common.

Current gain $\alpha < 1$. Very low input impedance ($\sim 50\Omega$) and very high output impedance ($\sim M\Omega$). No phase inversion (0°). Used in high-frequency and RF amplifiers, impedance matching from low to high impedance circuits.

2. Common Emitter (CE): Input at Base, output at Collector, Emitter is common.

Current gain $\beta = 20-500$ (highest). Medium input ($\sim 2k\Omega$) and output ($\sim 40k\Omega$) impedance. Output is 180° phase-inverted from input. Provides the highest power gain of all three configurations. Most widely used in audio amplifiers, switching circuits, and general-purpose amplifiers.

3. Common Collector (CC) / Emitter Follower: Input at Base, output at Emitter, Collector is common. Current gain $\approx \beta + 1$. Very high input impedance ($> 100k\Omega$) and very low output impedance ($\sim 50\Omega$). No phase inversion (0°). Voltage gain is less than 1 (output "follows" input). Used as a buffer amplifier and for impedance transformation from high to low impedance.

Key Relations:

$$I_E = I_B + I_C \quad | \quad \alpha = I_C/I_E \quad | \quad \beta = I_C/I_B \quad | \quad \beta = \alpha/(1-\alpha)$$

[Refer to Figs. 9.1, 9.2, 9.3 for configuration diagrams]

Q8. Explain the working of a CE amplifier with a neat diagram.**5 Marks****How does it achieve voltage amplification?****Common Emitter Amplifier – Working:**

The CE amplifier uses a transistor in CE configuration with biasing resistors R_1 , R_2 (voltage divider bias), a collector resistor R_C , emitter resistor R_E , bypass capacitor C_E , and coupling capacitors C_1 and C_2 .

DC Biasing (Q-Point Setting): R_1 and R_2 form a voltage divider that maintains a stable base voltage $V_B = V_{CC} \times R_2 / (R_1 + R_2)$. This sets the transistor operating point (Q-point) in the active region.

Input Stage: The AC input signal v_{in} is coupled through C_1 (which blocks DC) to the base terminal. This superimposes a small AC variation on the DC bias.

Amplification: The small base current variation ΔI_B causes a large collector current variation $\Delta I_C = \beta \cdot \Delta I_B$. This large current through R_C creates a large output voltage: $v_{out} = -\Delta I_C \times R_C$.

Phase Inversion: During the positive half of v_{in} , I_B and I_C increase $\rightarrow$ voltage drop across R_C increases $\rightarrow V_{CE}$ decreases $\rightarrow$ output goes negative. So the output is 180° out of phase with input.

Voltage Gain: $A_v = -\beta \cdot R_C / r_e$ where $r_e = 26\text{mV} / I_C$ (AC emitter resistance).

Output Coupling: C_2 couples only the AC amplified signal to the load, blocking DC.

[Refer to Fig. 11.1 for the complete CE amplifier circuit diagram]

Section 14: Quick Revision Sheet

All key formulas, definitions, and important points for last-minute revision

All Key Formulas

PN Diode

- $I = I_S(e^{V/\eta V_T} - 1)$
- $V_T = kT/q \approx 26\text{mV}$ (300K)
- $\eta = 1$ (Ge), $\eta = 2$ (Si)
- $V_{\text{cut-in}}$: 0.3V (Ge), 0.7V (Si)

Zener Diode

- $V_{\text{out}} = V_Z$ (constant)
- $I_S = (V_{\text{in}} - V_Z)/R_S$
- $I_S = I_Z + I_L$
- $I_L = V_Z/R_L$

BJT Current Relations

- $I_E = I_B + I_C$
- $\alpha = I_C/I_E$ (< 1)
- $\beta = I_C/I_B$ (20–500)
- $\beta = \alpha/(1-\alpha)$
- $\alpha = \beta/(\beta+1)$

CE Amplifier

- $A_v = -\beta R_C/r_e$
- $r_e = 26\text{mV}/I_C$
- $A_i = \beta$
- $A_p = A_v \times A_i$
- Phase shift = 180°

Key Points to Remember

Topic	Key Point
Semiconductor	Energy gap: Insulator $\sim 10\text{eV}$, Semiconductor $\sim 1\text{eV}$, Conductor $\sim 0\text{eV}$
N-type	Pentavalent dopant (P, As, Sb) $\rightarrow$ free electrons (majority carriers)
P-type	Trivalent dopant (B, Al, Ga) $\rightarrow$ holes (majority carriers)
Depletion region	No free charge carriers; has built-in electric field opposing diffusion
Forward bias	+ve to P, -ve to N; reduces depletion; low resistance; large current
Reverse bias	+ve to N, -ve to P; widens depletion; high resistance; tiny I_S
Zener effect	Tunneling in heavily doped junctions at $V_Z < 6\text{V}$

Avalanche effect	Carrier multiplication by impact ionization at $V_Z > 6V$
BJT Emitter	Heavily doped → injects carriers; most important terminal
BJT Base	Very thin & lightly doped → most injected carriers pass through to collector
BJT Collector	Moderately doped; largest region; collects carriers from base
Active region	E-B: forward biased; C-B: reverse biased → linear amplification
Saturation region	Both junctions forward biased → transistor acts as closed switch
Cutoff region	Both junctions reverse biased → transistor acts as open switch
CE Amplifier	180° phase inversion; highest power gain; most common configuration

⚡ Important Definitions (One-Line)

Device Definitions

- **Diode:** 2-terminal, unidirectional conductor
- **Zener:** Constant V regulator in reverse breakdown
- **BJT:** 3-terminal current-controlled device
- **Amplifier:** Signal magnitude increaser
- **Transistor:** Transfer-resistor (amplify/switch)

Configuration Quick-Ref

- **CB:** $\alpha < 1$, Z_{in} very low, Z_{out} very high
- **CE:** $\beta = 20-500$, highest power gain, 180° phase
- **CC:** $\gamma \approx \beta$, Z_{in} very high, Z_{out} very low

📊 Diagram Quick-Reference Summary



PN Diode
Depletion center



Zener Diode
 $V_{out} = V_Z(\text{const})$



NPN Symbol
 $\beta = I_C / I_B$

CE Amplifier
 $V_{in} \rightarrow \text{Base}$
 $V_{out} \leftarrow \text{Collector}$
180° Phase Inversion
 $A_v = -\beta R_C / r_e$
 Highest Power Gain

Fig. 14.1: Quick Reference – Key Device Symbols and Configurations

EXAM TIPS:

- Always draw neat, labeled diagrams — each diagram is worth 1–2 marks.
- Write all formulas before starting explanations in 5-mark answers.
- Remember: CE has 180° phase inversion; CB and CC have 0° (no inversion).
- For Zener regulator: always write $I_S = I_Z + I_L$ and $V_{out} = V_Z$.
- BJT terminal current: $I_E = I_B + I_C$ — never forget this KCL relation.
- Cut-in voltage: Silicon = 0.7V, Germanium = 0.3V — frequently asked.

BEEE – Unit IV | JNTUK R23 | Semester I

Semiconductor Devices Study Material | All Diagrams are SVG – Exam Ready

--- End of Unit IV ---